Harish G

AI / ML Engineer | Forecasting & Traditional ML in Production, GenAI, RAG, Agentic Systems

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Profile

I take models to production and I report what they actually do. On the traditional side that means gradient-boosted and global forecasting models measured out-of-sample against a mandatory seasonal-naive baseline, with conformal intervals derived from residuals and an accuracy certification that says plainly which forecasts clear the client's tolerance and which do not. On the generative side it means agents that call tools through declared schemas, retrieval that runs before inference with authorisation applied inside the retriever, and guardrails that mark any number the model did not get from a tool. Software Developer at CortexCraft.ai, in Python and Java.

Core Skills

Machine Learning: LightGBM quantile regression, PyCaret, scikit-learn, TensorFlow / Keras (LSTM classifiers), PyTorch; QLoRA fine-tuning with `transformers`, `peft`, `trl` and `bitsandbytes`; lag, rolling-window and calendar feature engineering; backtesting on holdout and rolling folds; WAPE, MASE, WMAPE, RMSE and R^2; conformal prediction intervals; champion selection against a baseline that ships if nothing beats it

GenAI & Agentic: LLM function and tool calling (runtime Gemini function declarations, bidirectional tool-call loop), config-driven tool registries and generic HTTP tool execution, MCP-style tool hosting (`inputSchema`, `readOnlyHint`, `destructiveHint`), role-based tool routing, multi-agent prompt design, numeric output guardrails

RAG & Vector Storage: pgvector (HNSW index, cosine distance), Gemini `text-embedding-004`, Hugging Face Inference (`all-MiniLM-L6-v2`, 384-d), chunking with overlap, Postgres full-text fallback, role-scoped row-level retrieval, Neo4j graph-enriched retrieval

MLOps & DevOps: model persistence and hot reload, retraining pipelines, Docker and Compose, four-stage GitLab CI with staging and production gates, Jenkins, GitHub Actions, Helm charts with per-environment values, Terraform for managed cloud services, Prometheus metrics, structured JSON logging with request-ID correlation

Languages & Backend: Python, Java 21, JavaScript / TypeScript, SQL; FastAPI, Spring Boot, Node / Express, Celery, Redis, Streamlit; PostgreSQL / pgvector, MongoDB, Neo4j; React

Experience

Software Developer, CortexCraft.ai Jan 2026 – Present

- Authored 380 of 418 commits on the voice-agent SaaS platform, behind 92 pytest suites, Playwright end-to-end tests and a four-stage GitLab CI pipeline.
- Built the agent tool layer: Gemini function declarations generated at runtime from agent configuration, a bidirectional tool-call loop and a generic HTTP tool executor, so onboarding a customer system is configuration rather than code.
- Built transcript document intelligence on Gemini JSON mode: schema-driven field extraction that coerces each value to its declared type and nulls anything off-type rather than fabricating it, plus summarisation with key points, sentiment and per-topic aspect tags.
- Own the deployment path end to end, as sole author of the deploy tree: three container images, a Helm chart with per-environment values, and Terraform for the managed services.

Freelance Web & App Developer (Self-Employed) 2023 – 2026

Delivered full-stack and data projects end to end for clients, then supported them through deployment, fixes and upgrades.

Selected Projects

Demand Forecasting System *Python, PyCaret, scikit-learn, Docker, Streamlit, Playwright*

- Sole author. Monthly forecasts across roughly 1,500 product series of 36 points each, from a tiered model zoo of about 150 labelled configurations: pure-numpy statistical models, a global ML tier, and a PyCaret sweep on aggregates.
- The ML tier fits one model across all series pooled rather than one per series, which is what makes a 48-configuration sweep affordable on 36-point histories, and forecasts the 12-month horizon directly rather than recursively, so no prediction is ever fed back as an input and error cannot compound across the horizon.
- Scored out-of-sample on WAPE and MASE rather than MAPE, which explodes on zero months, with seasonal-naive as a mandatory baseline that ships if no candidate beats it. Prediction intervals are conformal, computed from model residuals rather than from the variance of the data, and a GREEN / AMBER / RED certification reports that median product-level WAPE is 16.6% and 18.7% of series land inside the client's requested 10% band, rather than narrowing every interval until the requirement appears to be met.

Inventory & Sales Prediction *Python, LightGBM, FastAPI, PostgreSQL, Docker*

- Sole author. Three quantile LightGBM models give a lower, median and upper bound per product-store-day, over lag features at 1, 2, 3, 7, 14 and 28 days plus rolling mean, standard deviation, minimum and maximum windows and calendar features.
- Concept to production in one pipeline: train / validation / test split, WMAPE, MAE, RMSE and R^2 scoring, joblib persistence, a model loader that hot-reloads without a restart, a retrain entry point, and FastAPI serving in Docker.

Dental Clinic AI Assistant *Python, Flask, PostgreSQL/pgvector, Gemini, WhatsApp Business API*

- Chunked document RAG (800-token chunks, 100 overlap) embedded with `text-embedding-004` into pgvector, retrieved by cosine distance with a full-text fallback and role-scoped row filtering applied inside the retriever, so authorisation cannot be prompted around. Gemini function calling for booking behind a role-based tool router.

Zentrax, voice and gesture desktop assistant *Python, PyTorch, TensorFlow/Keras, transformers*

- QLoRA fine-tuning pipeline for SmolLM2 over a generated instruction dataset, with checkpoint resume and TensorBoard; separately, a Keras LSTM classifier and a scikit-learn random forest for gesture and voice-command recognition.

Education

B.E., Information Science and Engineering, Vivekananda Institute of Technology, Bangalore.